JEE–MAIN EXAMINATION – JANUARY 2026

(HELD ON THURSDAY 22ND JANUARY 2026) **TIME : 9:00 AM TO 12:00 NOON**

CHEMISTRY — TEST PAPER WITH ANSWER

SECTION-A

51. Consider the transition metal ions Mn^{3+}, Cr^{3+}, Fe^{3+} and Co^{3+} and all form low spin octahedral complexes. The correct decreasing order of unpaired electrons in their respective d-orbitals of the complexes is

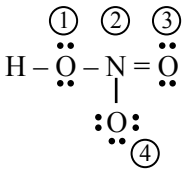
(1) $Cr^{3+} > Fe^{3+} > Co^{3+} > Mn^{3+}$

(2) $Mn^{3+} > Fe^{3+} > Co^{3+} > Cr^{3+}$

(3) $Fe^{3+} > Co^{3+} > Mn^{3+} > Cr^{3+}$

(4) $Cr^{3+} > Mn^{3+} > Fe^{3+} > Co^{3+}$

52. The formal changes on the atoms marked as (1) to (4) in the Lewis representation of HnO_3 molecule respectively are



(1) +1, 0, 0, –1 (2) 0, –1, 0, +1

(3) 0, +1, 0, –1 (4) 0, 0, –1, +1

53. Given below are two statements :

Statement I : Phenol on treatment with $CHCl_3$/aq. KOH under refluxing condition, followed by acidification produces *p*-hydroxy benzladehyde as the major product and *o*-hydroxy benzaldehyde as the minor product.

Statement II : The mixture of *p*-hydroxybenzaldehyde and *o*-hydroxybenzaldehyde can be easily separated through steam distillation.

In the light of the above statements, choose the ***correct*** answer from the options given below :

(1) Both Statement I and Statement II are false

(2) Statement I is true but Statement II is false

(3) Both Statement I and Statement II are true

(4) Statement I is false but Statement II is true

54. The energy required by electrons, present in the first Bohr orbit of hydrogen atom to be excited to second Bohr orbit is _____ J mol^{-1}.

Given : $R_H = 2.18 \times 10^{-11}$ ergs.

(1) 1.635×10^{-18}

(2) 9.835×10^{5}

(3) 9.835×10^{12}

(4) 1.635×10^{-11}

55. A 'p'-block element (E) and hydrogen form a binary cation $(EH_x)^+$, while EH_3 on treatment with K_2HgI_4 in alkaline medium gives a precipitate of basic mercury(II)amido-iodine. Given below are first ionisation enthalpy values (kJ mol^{-1}) for first element each from group 13, 14, 15 and 16. Identify the correct first ionisation enthalpy value for element E.

(1) 1312

(2) 1086

(3) 1402

(4) 801

56. In the reaction,

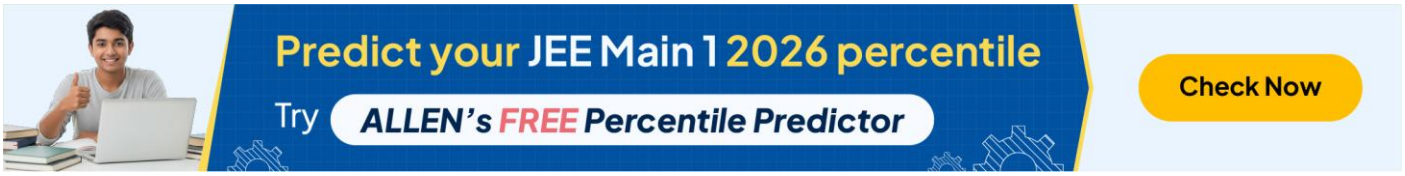
$2Al(s) + 6HCl(aq) \rightarrow 2Al^{3+}(aq) + 6Cl^{-}(aq) + 3H_2(g)$

(1) 11.2 L $H_2(g)$ at STP is produced for every mole of HCl consumed.

(2) 67.2 L $H_2(g)$ at STP is produced for every mole of Al that reacts.

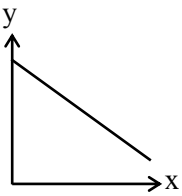
(3) 12 L HCl(aq) is consumed for every 6L $H_2(g)$ produced.

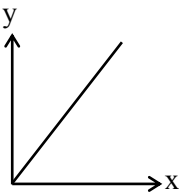
(4) 33.6 L $H_2(g)$ is produced regardless of temperature and pressure for every mole of Al that reacts.

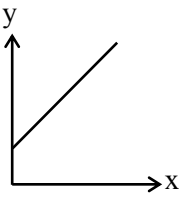


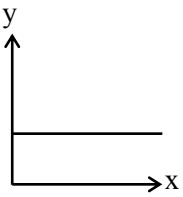
57. Consider a solution of $CO_2(g)$ dissolved in water in a closed container.

Which one of the following plots correctly represents variation of log (partial pressure of CO_2 in vapour phase above water) [y-axis] with log (mole fraction of CO_2 in water) [x-axis] at 25°C ?

(1) 

(2) 

(3) 

(4) 

58. A first row transition metla (M) does not liberate H_2 gas from dilute HCl. 1 mol of aqueous solution of MSO_4 is treated with excess of aqueous KCN and then $H_2S(g)$ is passed through the solution. The amount of MS (metal sulphide) formed from the above reaction is _____ mol.

(1) 2

(2) 1

(3) 3

(4) 0

59. The correct order of reactivity of CH_3Br in methanol with the following nucleophiles is

F^-, I^-, $C_2H_5O^-$ and $C_6H_5O^-$

(1) $I^- > C_6H_5O^- > F^- > C_2H_5O^-$

(2) $I^- > C_2H_5O^- > C_6H_5O^- > F^-$

(3) $I^- > C_2H_5O^- > F^- > C_6H_5O^-$

(4) $I^- > F^- > C_6H_5O^- > C_2H_5O^-$

60. Match the **LIST-I** with **LIST-II**

List-I Thermodynamic Process		**List-II Magnitude in kJ**	
A.	Work done in reversible, isothermal expansion of 2 mol of ideal gas from 2 dm^3 to 20 dm^3 at 300 K.	I.	4
B.	Work done in irreversible isothermal expansion of 1 mol ideal gas from 1 m^3 to 3 m^3 at 300 K against A constant pressure of 3kPa.	II.	11.5
C.	Change in internal energy for adiabatic expansion of a 1 mol ideal gas with change of temperature = 320 K and $\bar{C}_V = \frac{3}{2}R$.	III.	6
D.	Change in enthalpy at constant pressure of 1 mole ideal gas with change of temperature = 337 K and $\bar{C}_P = \frac{5}{2}R$.	IV.	7

Choose the ***correct*** answer from the option given below :

(1) A-III, B-II, C-IV, D-I

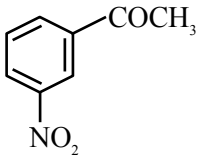
(2) A-II, B-III, C-I, D-IV

(3) A-I, B-II, C-III, D-IV

(4) A-II, B-I, C-III, D-IV



61. Given below are two statements :

Statement I : Benzene is nitrated to give nitrobenzene, which on further treatment with $CH_3COCl / AlCl_3$ will give 

Statement II : NO_2 group is a *m*-directing, and deactivating group.

In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Statement I is correct but Statement II is incorrect.

(2) Both Statement I and Statement II are correct.

(3) Statement I is incorrect but Statement II is correct.

(4) Both Statement I and Statement II is are incorrect.

62. A $\rightarrow$ products (First order reaction).

Three sets of experiment were performed for a reaction under similar experimental conditions.

Run 1 $\Rightarrow$ 100 mL of 10 M solution of reactant A

Run 2 $\Rightarrow$ 200 mL of 10 M solution of reactant A

Run 3 $\Rightarrow$ 100 mL of 10 M solution of reactant A+ 100 mL of H_2O added.

The correct variation of rate of reaction is

(1) Run 1 = Run 2 = Run 3

(2) Run 3 < Run 1 = Run 2

(3) Run 3 < Run 1 < Run 2

(4) Run 1 < Run 2 < Run 3

63. Match the LIST-I with LIST-II

List-I Reagents		**List-II Name of Reaction involving carbonyl compound**	
A.	NH_2-NH_2, KOH	I.	Tollen's Test
B.	$Ag(NH_3)_2OH$	II.	Clemmensen Reduction
C.	Aq. $CuSO_4$, Sodium Potassium tartarate, KOH	III.	Wolff-Kishner Reduction
D.	Zn – Hg, HCl	IV.	Fehling's Test

Choose the ***correct*** answer from the options given below

(1) A-III, B-I, C-IV, D-II

(2) A-II, B-I, C-IV, D-III

(3) A-IV, B-III, C-II, D-I

(4) A-III, B-IV, C-I, D-II

64. Given below are two statements :

Statement I : The halogen that makes longest bond with hydrogen in HX, has the smallest covalent radius in its group.

Statement II : A group 15 element's hydride EH_3 has the lowest boiling point among corresponding hydrides of other group 15 elements. The maximum covalency of that element E is 4.

In the light of the above statements, choose the ***correct*** answer from the options given below.

(1) Both Statement I and Statement II are true.

(2) Statement I is false but Statement II is true.

(3) Both Statement I and Statement II are false.

(4) Statement I is true but Statement II is false.



65. Given below are two statements:

Statement I : Sucrose is dextrorotatory. However sucrose upon hydrolysis gives a solution having mixture of products. This solution shows laevorotation.

Statement II : Hydrolysis of sucrose gives glucose and fructose. Since the laevorotation of glucose is more than the dextrorotation of fructose the resulting solution becomes laevorotatory.

In the light of the above statements, choose the correct answer from the options given below.

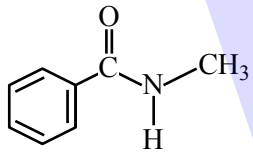
(1) Statement I is false but Statement II is true.

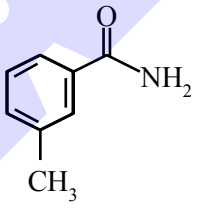
(2) Both Statement I and Statement II are false.

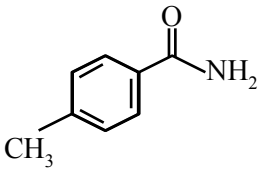
(3) Both Statement I and Statement II are true.

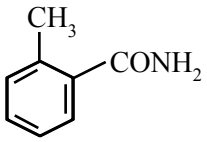
(4) Statement I is true but Statement II is false.

66. 'A' is a neutral organic compound (M. F : C_8H_9ON). On treatment with aqueous $Br_2/HO^{(-)}$, 'A' forms a compound 'B' which is soluble in dilute acid. 'B' on treatment with aqueous $NaNO_2/HCl(0–5°C)$ produces a compound 'C' which on treatment with CuCN/NaCN produces 'D' Hydrolysis of 'D' produces 'E' which is also obtainable from the hydrolysis of 'A'. 'E' on treatment with acidified $KMnO_4$ produces 'F'. 'F' contains two different types of hydrogen atoms. The structure of 'A' is

(1) 

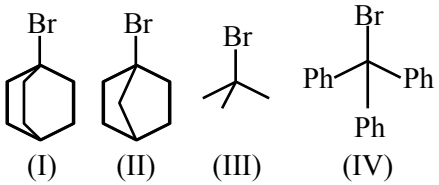
(2) 

(3) 

(4) 

67. The correct order of the rate of reaction of the following reactants with nucleophile by S_N1 mechanism is :

(Given : Structure I and II are rigid)



(I) (II) (III) (IV)

(1) IV < III < II < I (2) III < I < II < IV

(3) II < I< III < IV (4) I < II < III < IV

68. Two p-block elements X and Y form fluroides of the type EF_3. The fluoride compound XF_3 is a Lewis acid and YF_3 is a Lewis base. The hybridization of the central atoms of XF_3 and YF_3 respectively are

(1) Both sp^3 (2) sp^2 and sp^3

(3) sp^3 and sp^2 (4) Both sp^2

69. As compared with chlorocyclohexane, which of the following statements correctly apply to chlorobenzene ?

A. The magnitude of negative charge is more on chlorine atoms

B. The C – Cl bond has partial double bond character

C. C – Cl bond is less polar

D. C – Cl bond is longer due to repulsion between delocalised electrons of the aromatic ring and lone pairs of electrons of chlorine.

E. The C–Cl bond is formed using sp^2 hybridised orbital of carbon.

Choose the correct answer from the options given below :

(1) A, C and E only

(2) B, C and D only

(3) A, D and E only

(4) B, C and E only



70. Given below are two statements:

Statement I : The Henry's law constant K_H is constant with respect to variations in solution's concentration over the range for which the solutions is ideally dilute.

Statement II : K_H does not differ for the same solute in different solvents.

In the light of the above statements, choose the ***correct*** answer from the options.

(1) Statement I is false but Statement II is true.

(2) Statement I is true but Statement II is false.

(3) Both Statement I and Statement II are true.

(4) Both Statement I and Statement II are false.

SECTION-B

71. The cycloalkane (X) on bromination consumes one mole of bromine per mole of (X) and gives the product (Y) in which C:Br ratio is 3 : 1. The percentage of bromine in the product (Y) is _____ %. (Nearest integer)

(Given : Molar mass in g mol^{-1} H : 1, C : 12, O : 16, Br : 80)

72. Consider the following electrochemical cell at 298K

$$Pt\left|HSnO_2^-(aq)\right|Sn(OH)_6^{2-}(aq)\left|Bi_2O_3(s)\right|Bi(s).$$

If the reaction quotient at a given time is 10^6, then the cell EMF (E_{cell}) is _____ $\times$ 10^{-1}V (Nearest integer).

Given the standard half-cell reduction potential as

$E^0_{Bi_2O_3/Bi,OH^-} = -0.44V$ and

$E^0_{Sn(OH)_6^{2-}/HSnO_2^-,OH} = -0.90V$

73. The temperature at which the rate constants of the given below two gaseous reactions become equal is _____ K. (Nearest integer).

$X \longrightarrow Y \quad k_1 = 10^6 e^{\frac{-30000}{T}}$

$P \longrightarrow Q \quad k_2 = 10^4 e^{\frac{-24000}{T}}$

Given : ln 10 = 2.303

74. Sodium fusion extract of an organic compound (Y) with $CHCl_3$ and chlorine water gives violet color to the $CHCl_3$ layer. 0.15g of (Y) gave 0.12 g of the silver halide precipitate in Carius method. Percentage of halogen in the compound (Y) is _____ . (Nearest integer).

(Given : molar mass g mol^{-1} C : 12, H : 1, Cl : 35.5, Br : 80, I : 127)

75. Dissociation of a gas A_2 takes place according to the following chemical reactions. At equilibrium, the total pressure is 1 bar at 300K.

$$A_2(g) \rightleftharpoons 2A(g)$$

The standard Gibbs energy of formation of the involved substances has been provided below:

Substance	ΔG_t° / kJ mol^{-1}
A_2	–100.00
A	–50.832

The degree of dissociation of $A_2(g)$ is given by $(x \times 10^{-2})^{1/2}$ where x = _____.

(Nearest integer).

[Given : R = 8 J mol^{-1} K^{-1}, log 2 = 0.3010, log 3 = 0.48]



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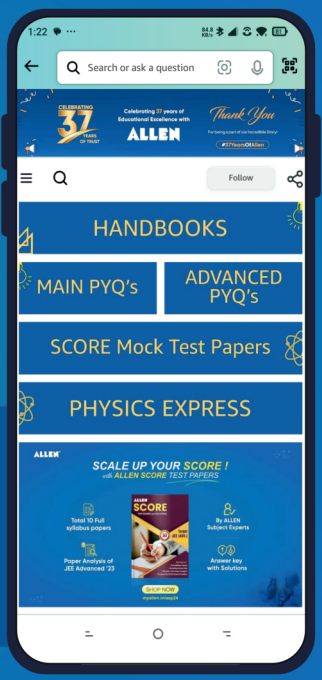
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